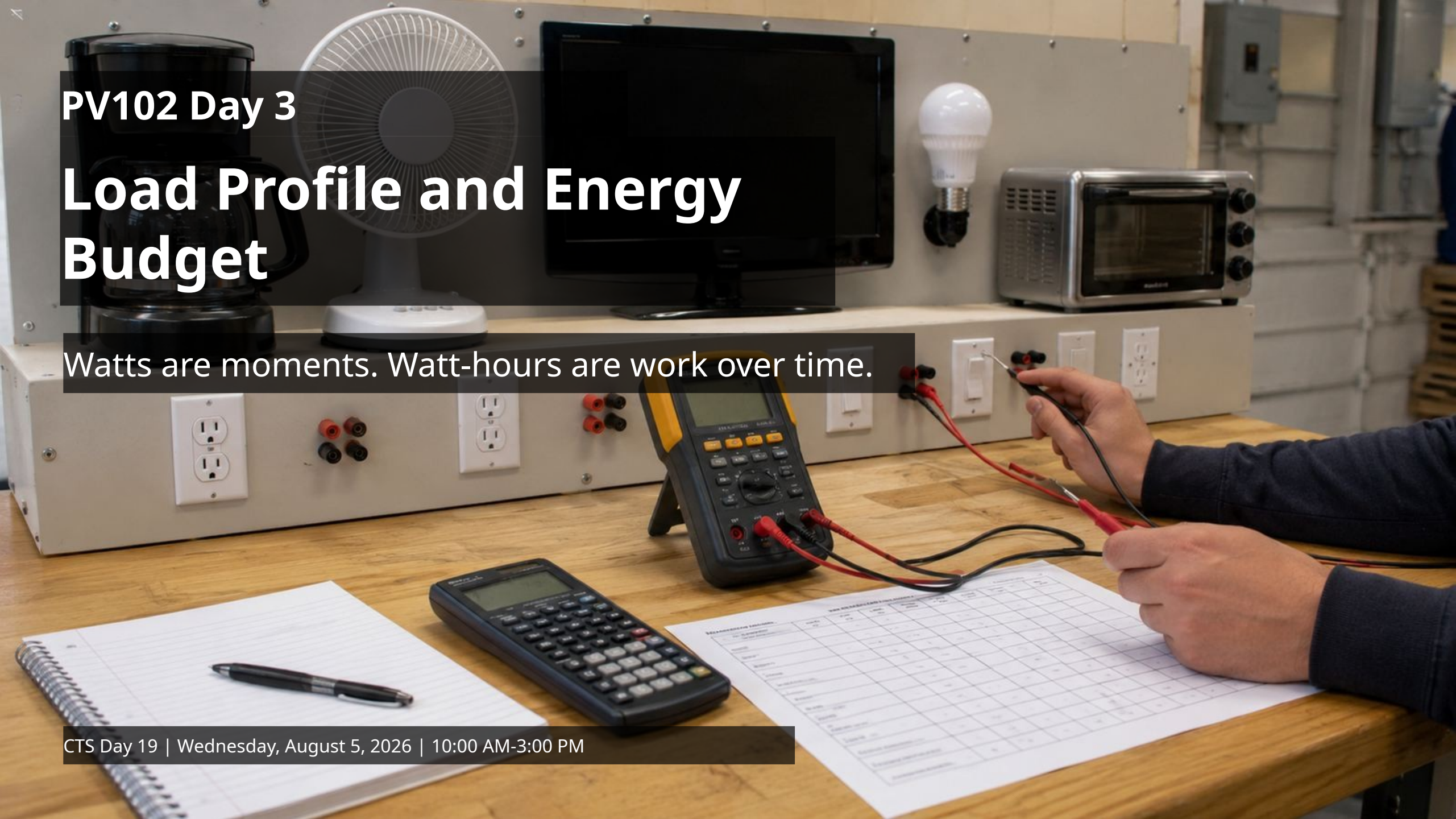
PV102 Day 3

Load Profile and Energy Budget

Watts are moments. Watt-hours are work over time.



CTS Day 19 | Wednesday, August 5, 2026 | 10:00 AM-3:00 PM

Today’s job

Professional solar work starts before parts move.

- Students turn a client wish list into a defensible load table, daily watt-hour estimate, priority list, and unknowns register.
- Carry forward the Day 1 habit: match architecture to real need.
- Today adds one design layer and one clean handoff.



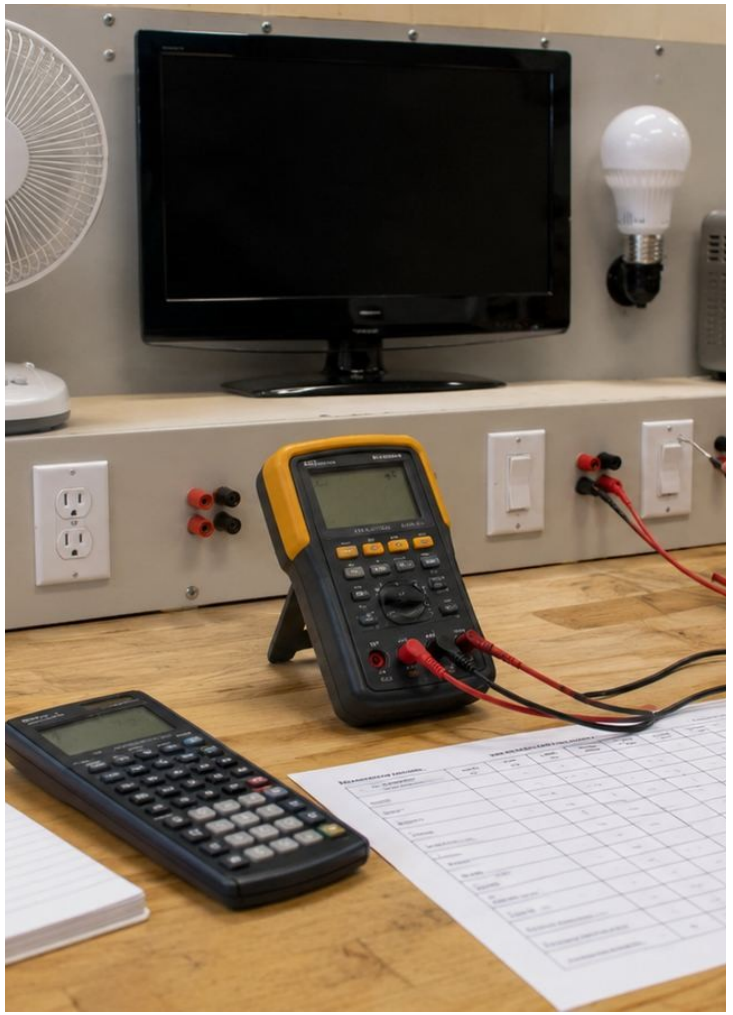
Do not guess when an unknown can be recorded.

Watts are moments. Watt-hours are work over time.

Learning targets

By 3:00 PM students should be able to:

- Distinguish watts from watt-hours.
- Calculate daily energy use from power and time.
- Rank loads by necessity, runtime, surge risk, and design impact.
- Explain how load decisions affect PV, battery, inverter, and budget.



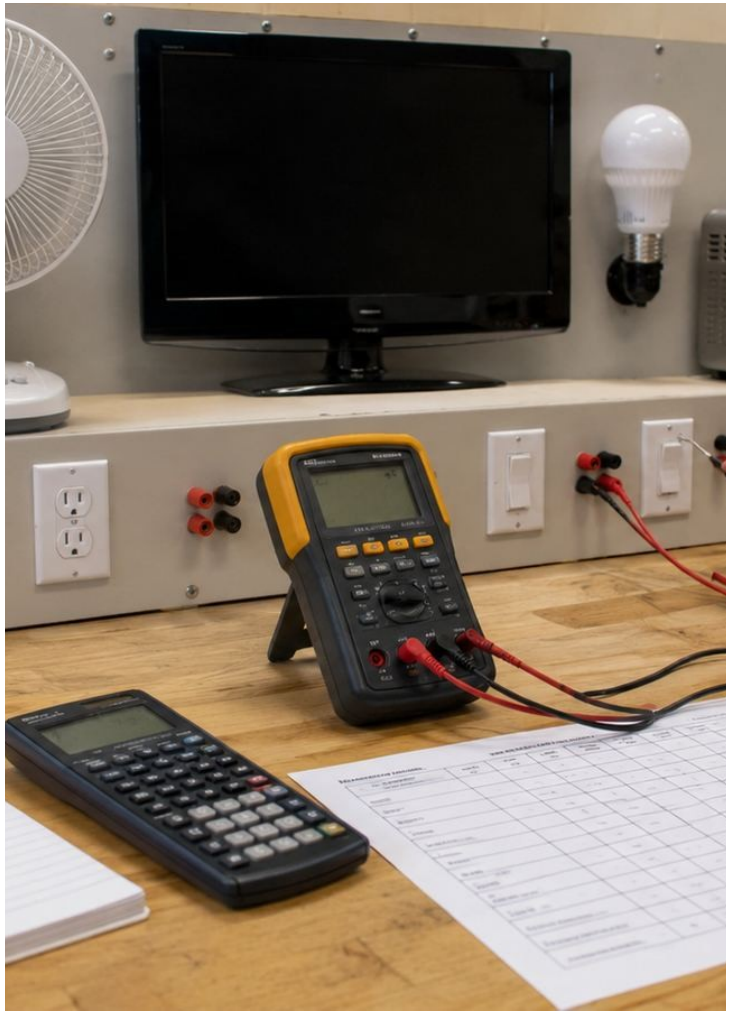
Evidence beats confidence. A tragic lesson for civilization, apparently.

Watts are moments. Watt-hours are work over time.

Morning readiness check

Quiz first, then review as evidence.

- Work independently for 12-15 minutes.
- Use wrong answers to locate teaching gaps.
- Ask: what fact would prove this answer?
- Keep review short and specific.

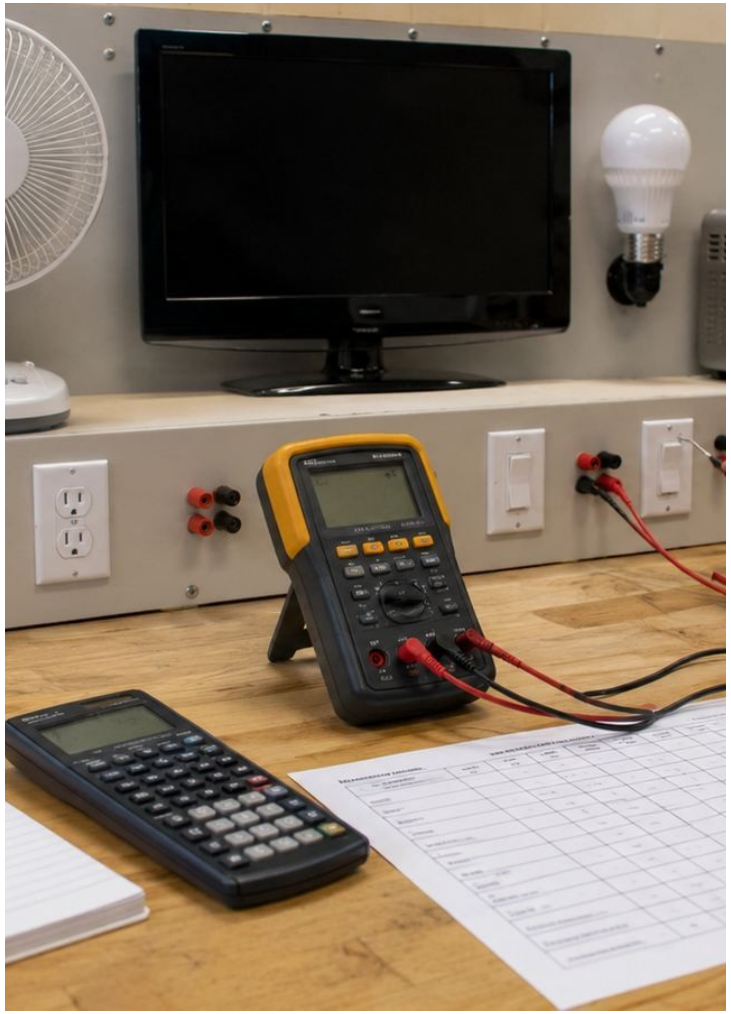


The quiz measures instruction as much as students.

Watts are moments. Watt-hours are work over time.

Core rule

- Watts are moments. Watt-hours are work over time.
- Say it once at the start.
- Bring every debate back to it.
- Use it to reject weak handoffs.



One rule, repeated until reality behaves.

Watts are moments. Watt-hours are work over time.

What students already know

PV101 and RCS are not separate islands.

- PV modules produce DC power.
- Series and parallel change voltage/current behavior.
- Continuity requires de-energized verification.
- Troubleshooting starts from evidence.
- Handoffs protect the next person.



Watts are moments. Watt-hours are work over time.

New judgment today

The skill moves from “can I build it?” to “should this be the design?”

- Define the job.
- Gather the evidence.
- Name the constraint.
- Protect the boundary.
- Record unknowns.



Watts are moments. Watt-hours are work over time.

Energy path and decision boundary



Every arrow is a boundary. Boundaries need ratings, protection, isolation, and documentation.

Watts are moments. Watt-hours are work over time.

Decision checks

Decision	Evidence	Reject if
Keep a design choice	Measured or observed site/load data	It only sounds good
Reject a shortcut	Safety or equipment boundary	It hides a hazard
Carry unknown forward	Named verification step	It becomes fake certainty
Approve handoff	Next person can act safely	It depends on mind reading

Guided example

Model one complete thinking path.

- Start with the client/job statement.
- Add one measured or observed fact.
- Make one design decision.
- Name one safety boundary.
- Name one unknown and how to verify it.

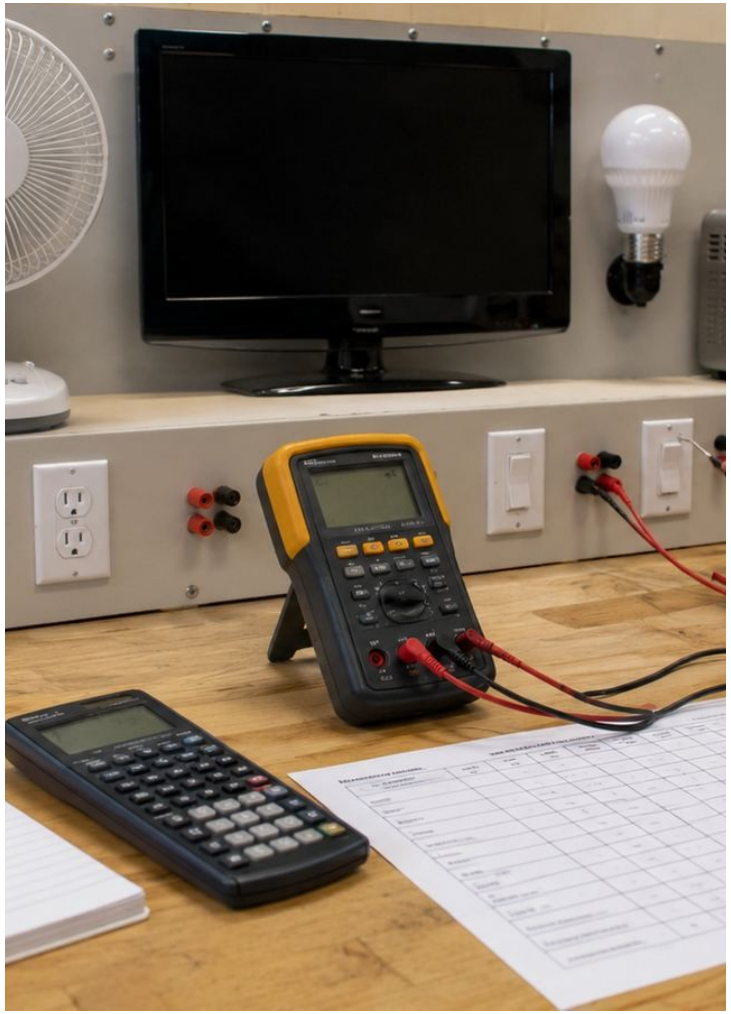


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Team roles

Three teams of two, roles rotate after the first pass.

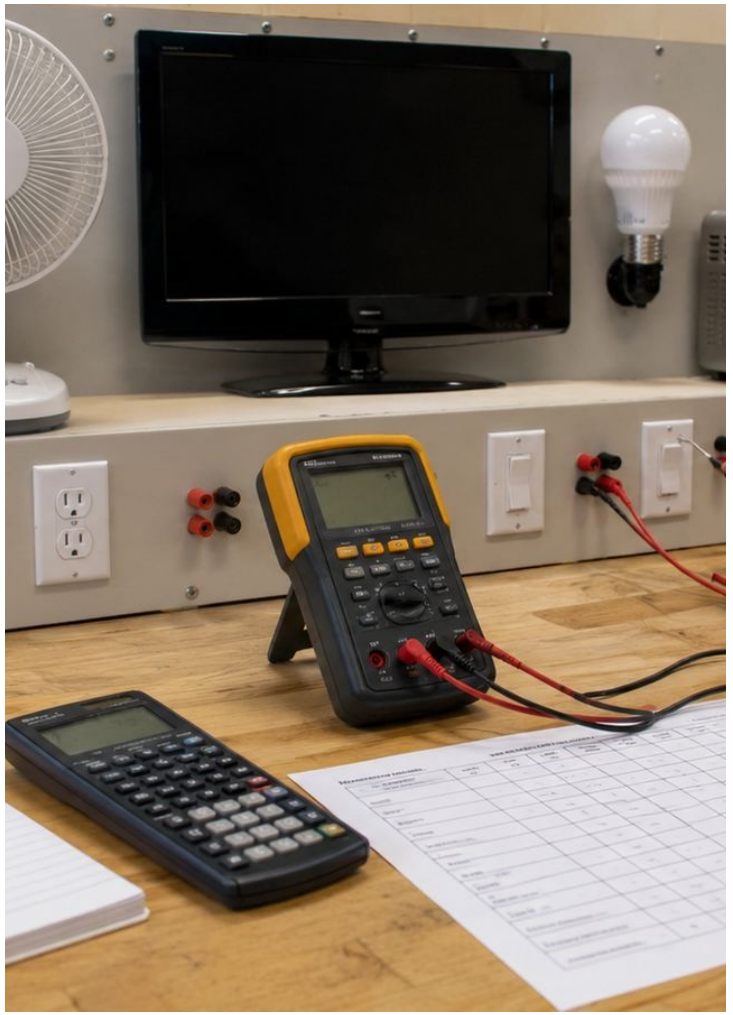
- Designer: proposes the first path.
- Verifier: challenges evidence and limits.
- Recorder/handoff lead: writes what the next person needs.
- In a pair, one person may hold two roles, then swap.



Lab mission

Client Load Interview and Energy Budget

- Teams interview a mock client, build a daily load table, and recommend one load reduction.



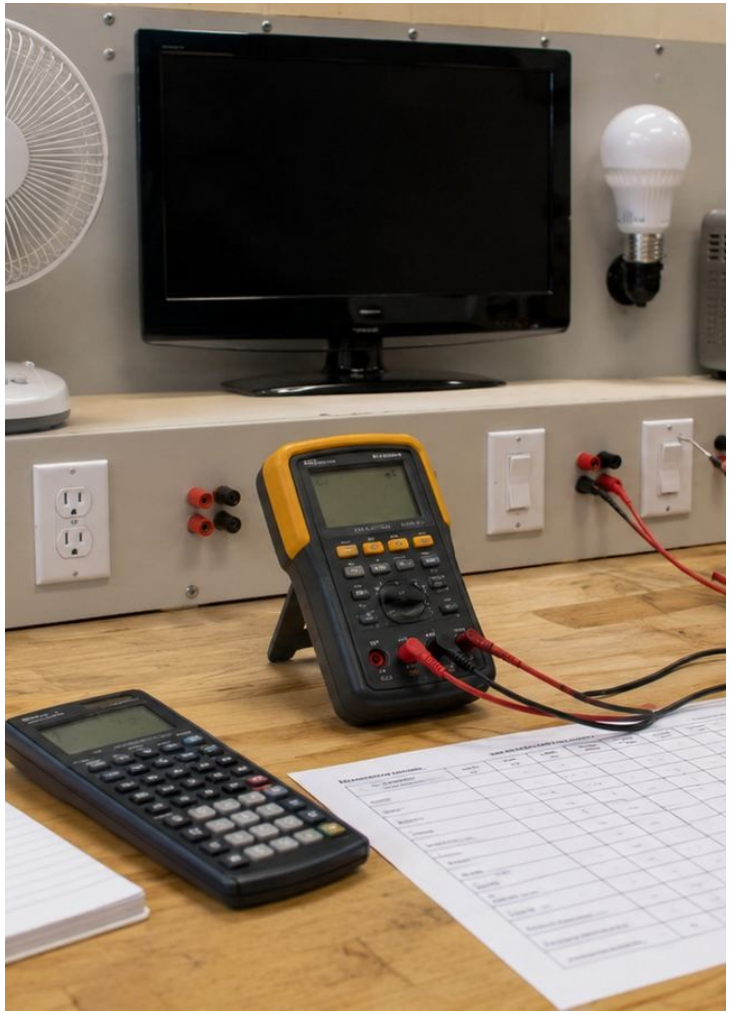
This is the work block. Keep it practical and evidence-based.

Watts are moments. Watt-hours are work over time.

Scenario options

Assign one, or let teams choose if the room can handle democracy today.

- Tiny shop: lights, laptop, phone charging, modem/router, small fridge, occasional circular saw.
- Remote trail counter: sensor, camera, cellular modem, winter operation, limited service visits.
- Bus conversion: lighting, fridge, fan, water pump, laptop, phone charging, occasional induction cooktop.

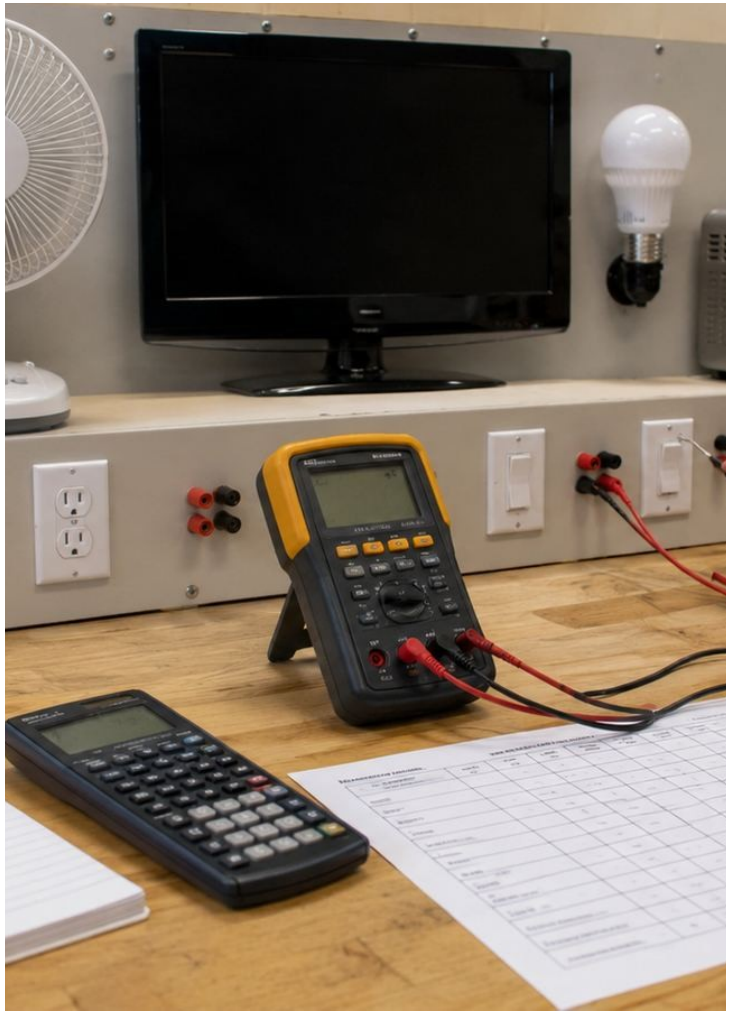


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Lab procedure

Run the same loop every day.

- Choose one scenario and list all loads.
- For each load, record watts, hours per day, daily Wh, priority, and unknowns.
- Mark surge or motor loads.
- Calculate total daily Wh and essential daily Wh.
- Recommend one load change that improves system practicality without weakening the client need.



Watts are moments. Watt-hours are work over time.

Acceptance criteria

A complete answer has five parts.

- Match the need.
- Show the path.
- Name the dominant constraint.
- State the safety boundary.
- Record unknowns instead of guessing.

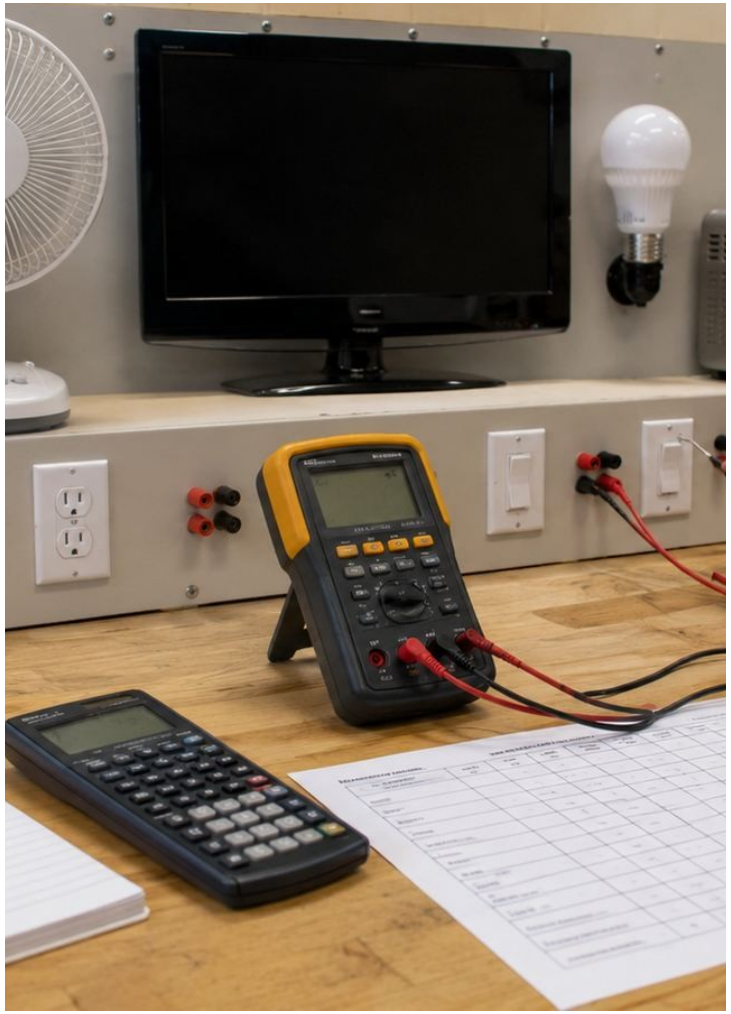


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Safety boundary

No training objective is improved by pretending energy is harmless.

- Use only instructor-approved equipment and procedures.
- Treat PV and batteries as energy sources.
- Stop when measurements do not make sense.
- Follow manufacturer instructions and program safety procedures.



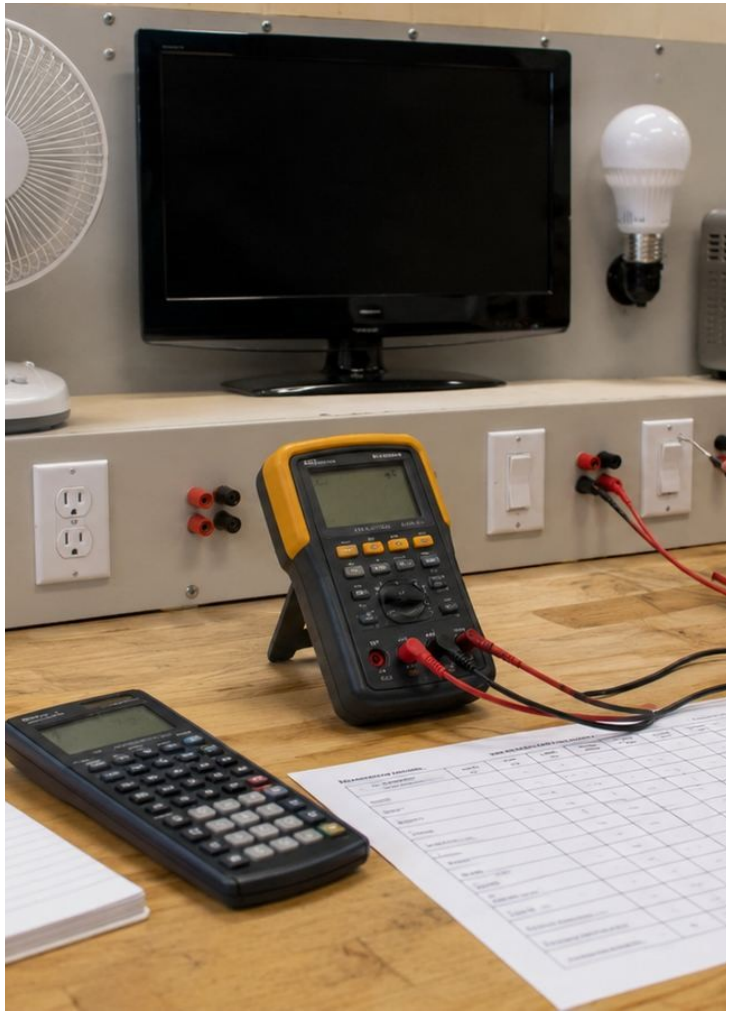
Stop, isolate, verify, ask. The glamorous life.

Watts are moments. Watt-hours are work over time.

Instructor probe questions

Use these when teams get vague.

- What evidence supports that?
- What would make this design fail?
- Which boundary protects the person or equipment?
- What unknown are you carrying forward?
- What would the next crew need to know?

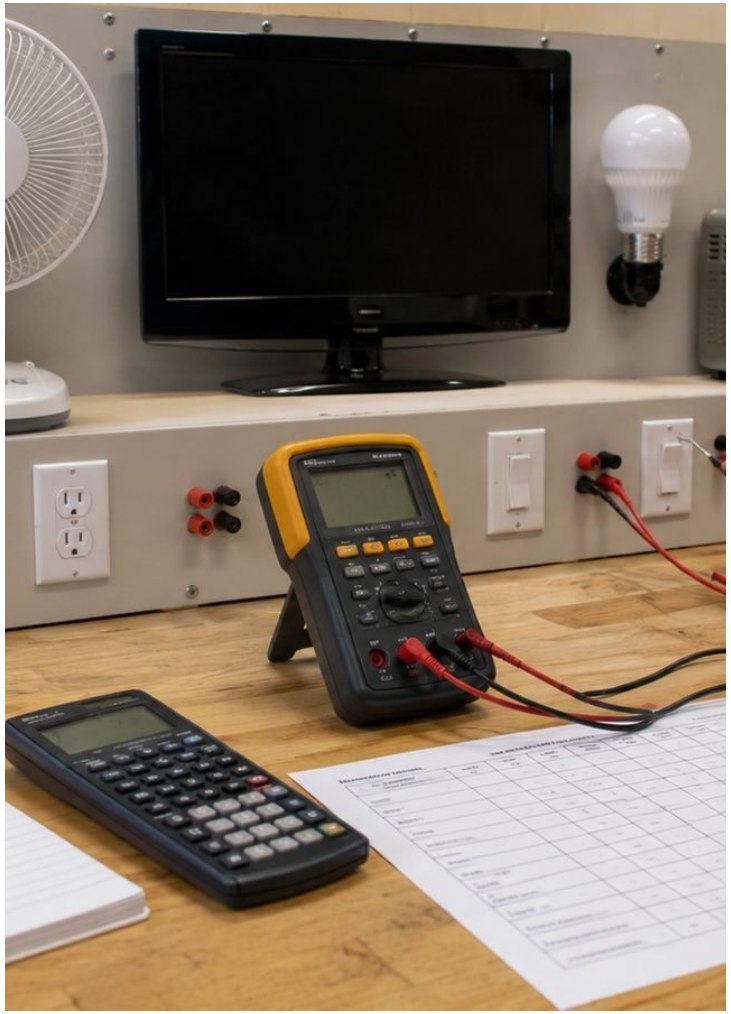


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Peer verification

Peers check the handoff, not the personality of the speaker. Humanity struggles here.

- Can I identify the job?
- Can I follow the path?
- Can I name the constraint?
- Can I see the safety boundary?
- Can I find the unknowns?

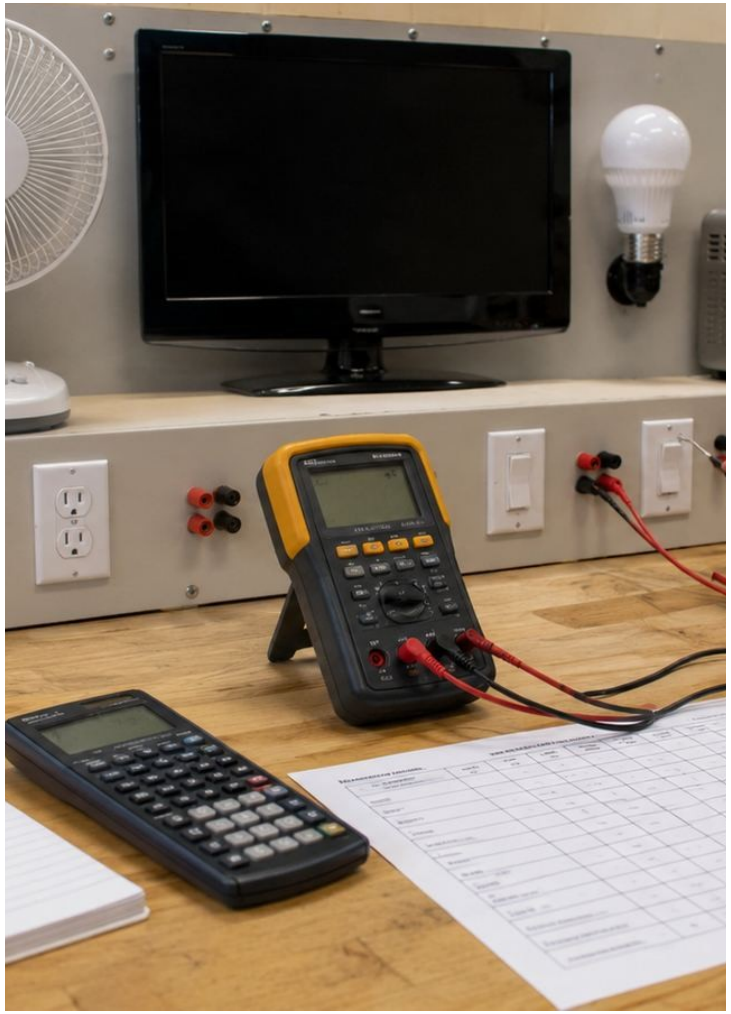


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Common failure modes

Correct these early.

- Component list with no job.
- Guess presented as fact.
- Missing safety boundary.
- No units or runtime.
- Handoff too vague to use.

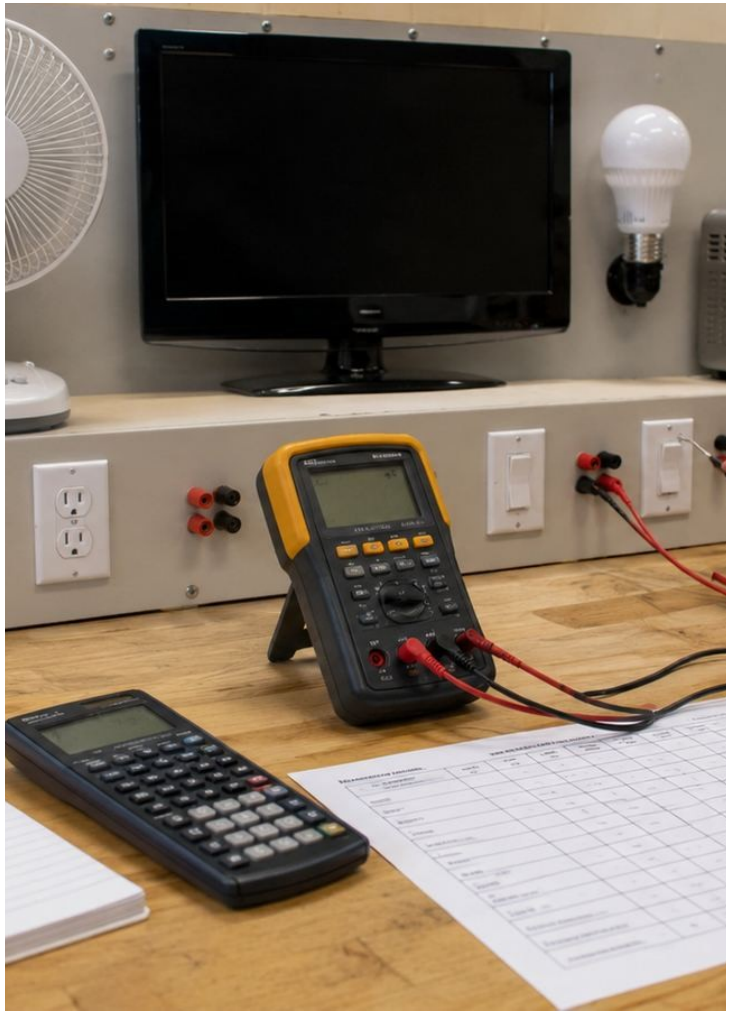


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Evidence language

Require technical sentences.

- Observed: what you saw.
- Measured: value with unit or clear method.
- Assumed: guess being carried forward.
- Verified: checked against source, procedure, or measurement.



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Handoff structure

Make the next person safer.

- Scope: what job is being solved.
- Decision: what was chosen or rejected.
- Evidence: why.
- Boundary: what must be protected.
- Unknowns: what still needs verification.

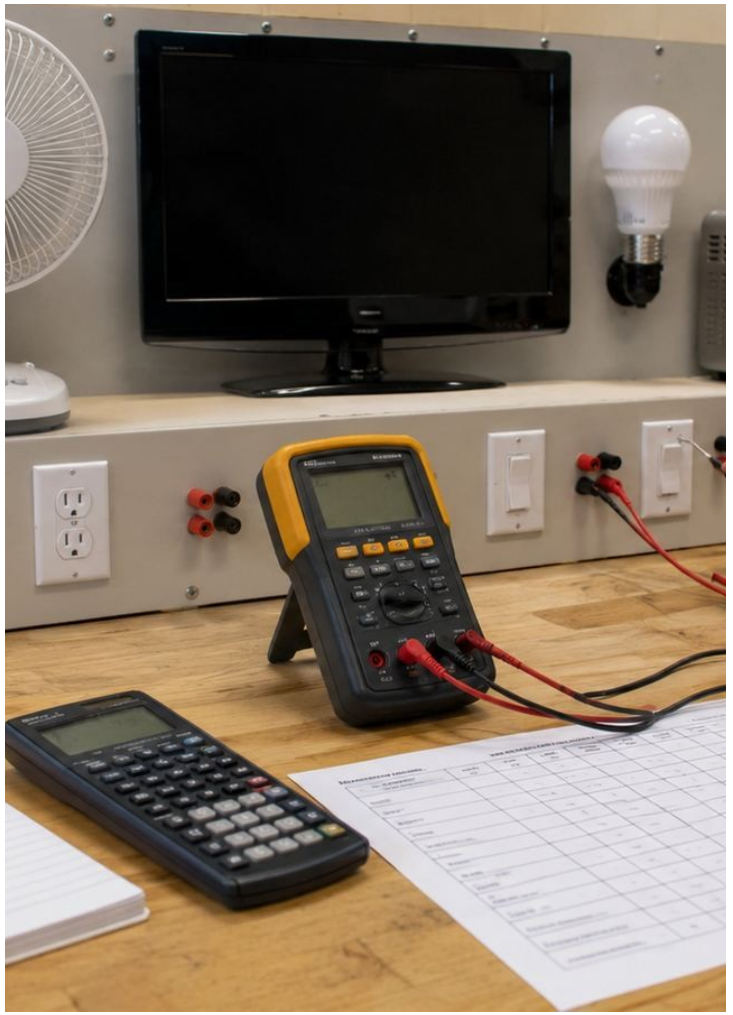


Watts are moments. Watt-hours are work over time.

Exit ticket

Individual close-out

- Which load drove the design hardest today: highest watts, longest runtime, or highest priority? Explain.



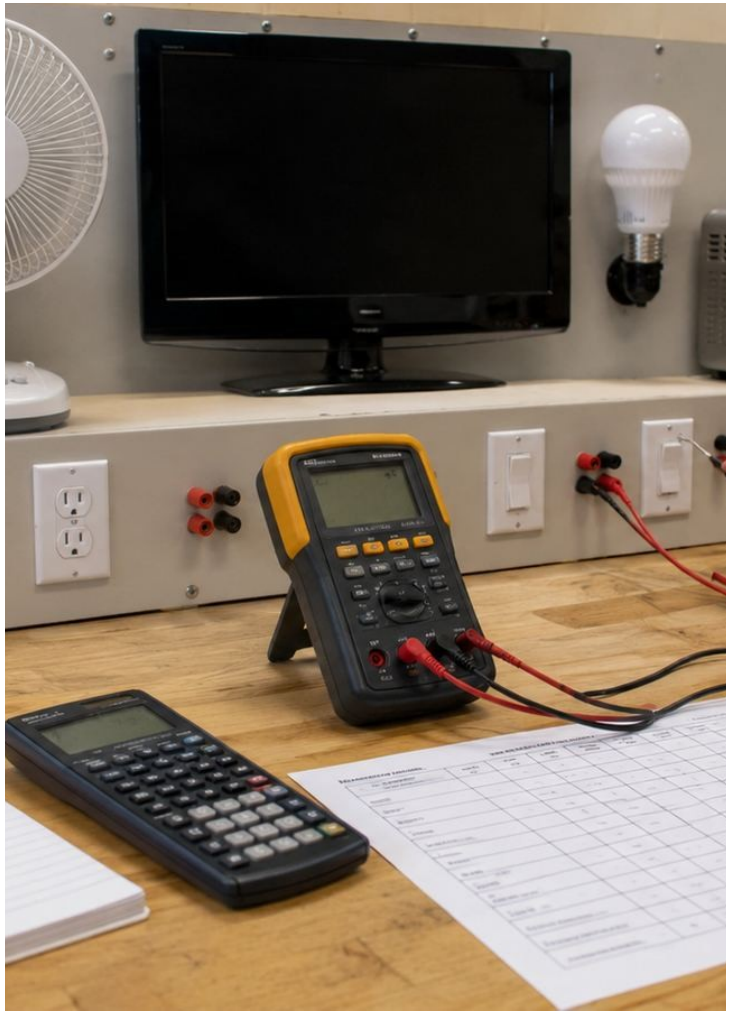
Collect before reset so the day leaves a paper trail. Weirdly useful.

Watts are moments. Watt-hours are work over time.

Day-to-day continuity

This is one system, not seven disconnected worksheets.

- PV102 Day 4 uses load and site information to explore module behavior, strings, and shade impact.
- Carry forward site notes, load tables, diagrams, and unknowns.
- Do not erase uncertainty. Resolve it.



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Final reminder

What professionals do

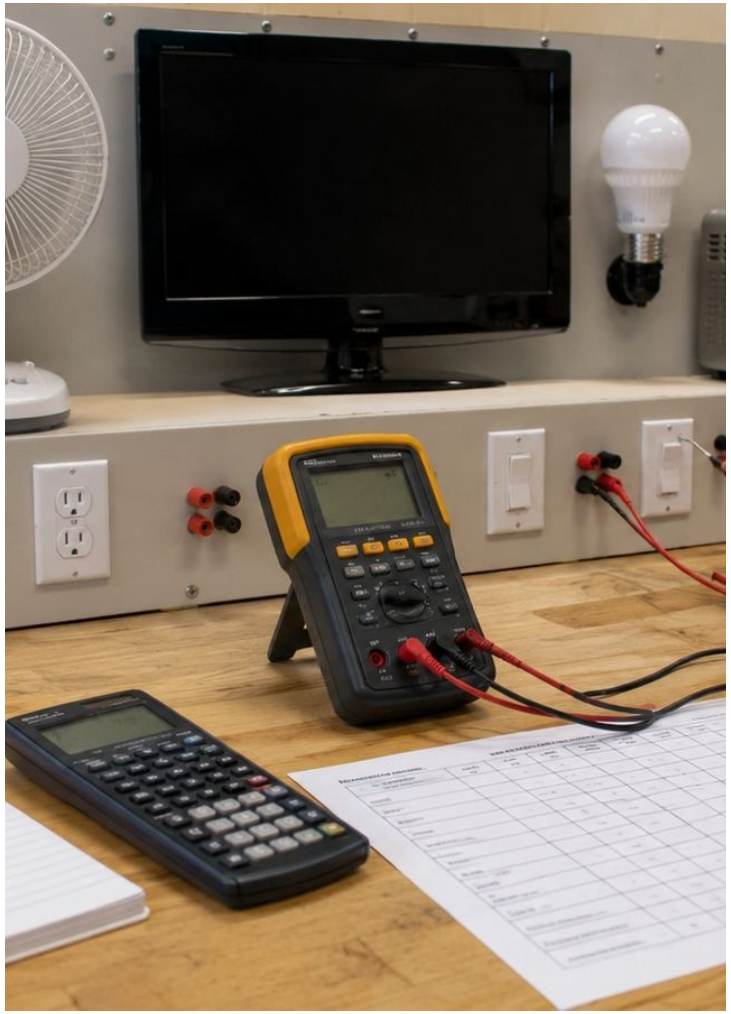
- Define the job.
- Verify the path.
- Protect the boundary.
- Document the truth.



Watts are moments. Watt-hours are work over time.

Close

- Watts are moments. Watt-hours are work over time.



Reset tools. Collect packets. Archive the handoffs.